

CONTENT SCRIPT + OFFSCREEN WORKER (extraction)

1 Descriptor extractor

DOM → text, colour, behaviour, readability (86 ms/page)

2 Sentence embedding

MiniLM, 384-d, in-browser ONNX

search

upsert

IndexedDB

cosine search → revisit flag; upsert this page

page descriptor

SERVICE WORKER (classifier cascade)

sensitive /
payment /
chrome://?

yes

Silence / bypass

no request leaves the device

no

3a Keyword

cat.: text only
mood: text+colour
+behaviour bias
stop if conf ≥ 0.5

escalate

3b Zero-shot NLI

abstains unless
score ≥ 0.45 ,
margin ≥ 0.10
hosted/local ONNX

abstain

3c Remote LLM

8 s timeout
falls back to 3a
on failure

page text

Entailment API
hosted, via key proxy

page text

LLM API
hosted, via key proxy

4 Confidence window: mood holds 5 s → music profile → prompt

generation request

GENERATION SERVICE (HTTP, off-device)

5 Validate profile

cache key: mood/style/key
+ bucketed fields

lookup

Postgres
audio_cache

hit

Return audio URL

+ loop point + metadata

miss

6 MusicGen-small

batching worker
(cold: minutes on CPU)

7 Loop synthesis

self-similarity cut, 50 ms
crossfade, -18 LUFS, Opus

save

Files
object store
(.ogg clips)

URL

fail

fail

8 Any stage fails → per-mood fallback clip (11 clips);
also served first, instantly, on every request

fallback clip



OFFSCREEN DOCUMENT (playback)

9a Fallback clip

audible in ~7 ms

9b Generated clip crossfades in

abort-on-stale: new mood cancels it

Player: two decks, gapless loop point → gain / effects



generated clip + loop point

— control / data flow

..... terminal stage

- - - - - page text leaves device

○ persistent store